WeatherNet

A 1-Billion-Parameter Ternary Neural Network for 24-Hour Station-Temperature Forecasting

Technical report | September 8, 2026 | v0

Abstract

WeatherNet is a compact transformer-style forecaster that maps a 24-hour window of four near-surface meteorological variables at a station to the next 24 hourly temperatures. The model contains **954,605,080 parameters (~0.955B)**, of which 99.97% live in 1.58-bit ternary linear layers and only 277,784 parameters -- the input/output heads -- are floating point. Training runs end-to-end on a single consumer 8 GB AMD GPU with a hybrid CPU/GPU optimizer and BF16 activations, at roughly 47-48 samples/s with batch 64. On a hold-out set of 3,000 windows (stations/windows never trained on) the model achieves an RMSE of **4.72 C** and an average hourly correlation of **0.887**, against 11.88 C for climatology and 5.20 C for persistence -- a **60.27% skill** improvement over the climatological mean (MAE basis 64.67%, pooled R^2 78.6%) and 9.24% over persistence, improving at every lead hour up to +24 h. Scaling evidence on 12k vs 300k training windows shows accuracy improving with data; a 1M-window run is in progress.

Keywords: numerical weather emulation, generation-transformer, ternary quantization, 1.58-bit networks, low-VRAM training, station-level temperature forecasting.

1 Introduction and motivation

Operational weather forecasting rests on numerical weather prediction (NWP) -- enormous physics simulations run on supercomputers. A parallel line of work treats forecasting as **statistical emulation**: learning the input/output map of the atmosphere directly from historical observations. Deep emulators such as FourCastNet, Pangu-Weather, Aurora and GenCast are now competitive with or better than operational NWP at meso-to-global scales, but they are trained with hundreds of GPUs and petabytes of reanalysis.

This project asks a different question: **how much skill can one person obtain from one available GPU and the free public observational record?** The result is WeatherNet -- a purposely over-parameterized (0.955B) but aggressively quantized network whose entire weight mass is stored as ternary {-1,0,+1} values. Ternary weights cut optimizer memory and bandwidth by more than an order of magnitude, letting a wide/deep network fit comfortably in an 8 GB budget while the floating-point effort concentrates in two thin input/output heads.

2 Data

We use the NOAA Global Historical Climatology Network daily summaries (GSOD): **26,627 stations** world-wide, converted to an hourly series of four near-surface variables -- temperature, pressure, humidity and wind speed. A sliding 24-hour window predicts the next 24 hourly temperatures (out_features = 24) using those four variables (24 x 4 = 96 inputs) plus the 24-hour mean of each variable (4 context features). Gaps in a station record are handled by restarting the window after each NaN, so every sample is fully observed. The full archive is estimated to contain **~8.6M windows**; we trained on 12k, then 300k, and are now running 1M.

3 Architecture

WeatherNet is a feed-forward network with a fixed-width ternary core (table 1). An FP32 input head expands the 100-dim context into a 1536-dim learned code, followed by **14 residual blocks** width 8192 -- all of them ternary BitLinear layers -- converging to a 24-dim FP32 head. The workhorse layer quantizes each weight to {-1,0,+1} with straight-through-estimator gradients: a dead-zone threshold of 0.1 x mean|W| handles the three-level split (93.6% of ternary weights are active +/-1 after training), and activations are quantized to an 8-bit grid. All quantized matmuls are sliced into chunks of 512 columns to keep peak activations low enough for an 8 GB card.

total parameters	954,605,080 (~0.955B)
ternary (BitLinear) weights	954,327,296 (99.97% of params)
FP32 head parameters	277,784
input head width / embedding	1536
hidden width / residual blocks	8192 / 14
projection width before FP32 head	256
quantization	ternary weights (thr 0.1), 8-bit activations
estimated FLOPs / sample (matmul)	1.91e9
VRAM used (batch 64)	~3.9 GB

Table 1. WeatherNet size and quantization budget.

4 Training procedure

Targets are normalized by a single (pooled) mean/std fitted on the **training** windows only. Loss is a Huber smooth-L1 on the 24-hour target. Optimizer is Adafactor in factored form (saving ~99% of Adam-style second-moment memory on the ternary weights). Two Adafactor instances run **in parallel on different hardware**: one on the GPU for the 954.2M ternary weights, one on the CPU in float32 for the 400,920 small 1-D parameters (biases, RMS gains, head). This hybrid dispatch keeps ROCm's Adafactor stable -- its 1-D path had a known NaN instability at 1e-30 epsilon on this stack; with eps=(1e-16, 1e-3) and 1-D params moved to CPU the training is NaN-free from step zero. Weights are stored BF16 on device; activations run under BF16 autocast.

Each epoch draws a random permutation of the streamed windows. Batches are 64 with batch-accumulation 1. On the AMD RX 6600 (8 GB, ROCm gfx1032) with HSA_OVERRIDE_GFX_VERSION=10.3.0 the steady-state rate is **47-48 samples/s** with the GPU ~99% busy and the CPU never the bottleneck. Epoch-0 checkpoint files are written atomically every 2,000 optimizer steps with full optimizer state, and training auto-resumes them -- a crash/reboot-safe protocol used operationally.

5 Experiments and results

Hold-out protocol: the **last 3,000 windows** of the canonical archive order (in effect later-period records of the newest stations) are reserved; the normalizer is fitted on the remaining 297,000 only. Baselines are computed on the **same** windows: climatology (pooled mean temperature, 8.70 C) and persistence (today's most recent temperature held for all 24 hours). The held-out carries a temperature standard deviation of 10.21 C, so samples span many climates.

method	RMSE (C)	MAE (C)	hourly corr	anomaly corr
climatology	11.88	10.10	-	0.000
persistence	5.20	3.85	-	0.869
WeatherNet (300k)	4.72	3.57	0.887	0.887

Table 2. 24-h-ahead station temperature, 3,000 window hold-out (identical window set for all methods).

lead hour	climo	persist	WeatherNet
+6 h	11.92	4.82	4.36
+12 h	11.86	5.17	4.71
+18 h	11.85	5.58	5.04
+24 h	11.84	5.96	5.41

Table 3. RMSE (C) by lead hour. WeatherNet beats persistence at every horizon, and its advantage grows deeper into the window -- the property that matters for planning use cases.

Scaling behaviour. At the same hold-out protocol the 12k-window model reached RMSE 5.81 C / corr 0.826. Going to 300k improved RMSE to 4.72 C / corr 0.887 (+18.8% RMSE, +6.1% corr). A 1M-window run is in progress; the full 8.6M archive is the next stage.

6 Limitations

- Target resolution is daily-step (24 h max); precipitation and other variables are not yet predicted.
- The hold-out contains an important but imperfect generalization test: windows are disjoint, and the split is effectively one of time period. A geographic station split is future work.
- Baselines shown are climatology and persistence. The decisive industry benchmark -- skill vs free NWP/GFS internet forecasts on the same stations -- is not yet computed.
- Outputs are deterministic point forecasts; probabilistic (ensemble) outputs are planned.
- Training-time target normalization is a single pooled scalar; per-hour or per-station normalization is expected to improve numbers further.
- This is not an operational product and in particular not financial advice.

7 Reproducibility

Single node: AMD RX 6600 8 GB, ROCm PyTorch 2.x Python 3.14, Ubuntu Linux. Required env: HSA_OVERRIDE_GFX_VERSION=10.3.0, PYTORCH_CUDA_ALLOC_CONF=expandable_segments:True. Source: weather1b repo (data.py, layers.py, model.py, train.py). Adafactor lr 0.01, eps=(1e-16,1e-3), grad-clip 1.0, BF16 autocast, chunk-size 512, ternary threshold 0.1, activation bits 8, seed fixed, batch 64.

8 Future work

- full-archive training (all ~8.6M windows);
- hourly-native variable (wind, solar irradiance, precipitation);
- probabilistic outputs via heteroscedastic or ensemble heads;
- skill comparison against free NWP/GFS baselines on identical stations;
- geographic/leave-climate-out validation, regional normalization.

All numbers in this report are reproducible from the repository checkpoints runs/weather-300k/final.pt (evaluation script baselines.py). Version 2026-09-08.